Aesthetic Evaluation of Facial Portraits Using Compositional Augmentation for Deep CNNs

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Abstract. Digital facial portrait photographs make up a massive portion of photos in the web. A number of methods for evaluating the aesthetics of photographs have been proposed recently. However, there have been a little work in the research community to address the aesthetics of targeted image domain, such as portraits. This paper introduces a new compositional-based augmentation scheme for aesthetic evaluation of portraits by well-known deep convolutional neural network (DCNN) models. We present a set of feature augmentation methods that take into account compositional photographic rules to ensure that the aesthetic in portraits are not hindered by standard transformations used for DCNN models. On a portrait subset of the large-scale AVA dataset, the proposed approach demonstrated a reasonable improvement in classification performance over the baseline and vanilla deep learning approaches.

1 Introduction

The ubiquity of smartphones and popularity of social media networks have contributed to an increasing number of photographs, particularly self-portraits or "selfies". A recent study by Bakhshi et al. [1] showed that photos with faces are more popular than other types of pictures. Nevertheless, casual photographers tend to produce many unprofessionally taken photos that are less aesthetically pleasing, and proceed to share them with other people in their circles. As such, an automatic assessment of the photo aesthetics would be useful to provide advice as to which of these photos should be retaken. This is also helpful for designers to select aesthetically pleasing photos for magazine covers, books, and news articles. However, different type of photos needs different type of photographic rules. Even, the most applicable rules, such as rule of thirds [2] could be applied differently for portraits and other type of photos [3].

Deep learning have been widely used in computer vision and natural language processing fields in recent years. In particular, deep convolutional neural network (DCNN) models have made revolutionary steps in object detection and recognition tasks [4–6]. The usage of graphic processing units (GPUs) have made it possible to handle large amounts of data, which directly provides better generalization and discrimination of models. Despite these technological advances and a growing research interest in computational aesthetics ever since the work

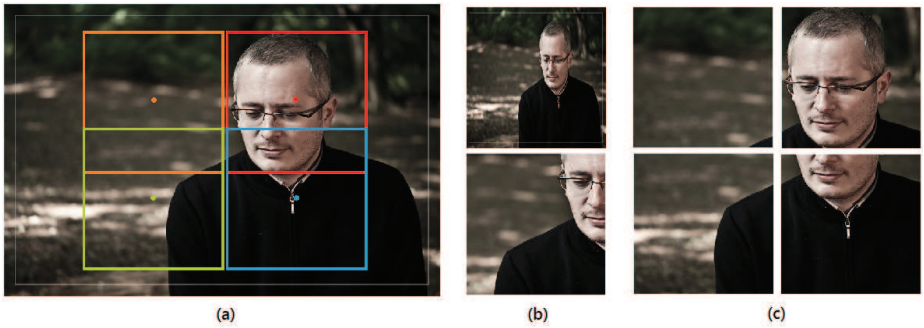

Fig. 1. Compositional-based cropping for DCNN inputs: (a) A portrait can be cropped aesthetically from power points defined by rule of thirds composition principle, (b) Full image warping *(top)* and center-cropping *(bottom)* can typically distort composition in portraits. (c) The four sub-images based on the power points.

of Datta et al. [7], very little work has been done in the specific domain of facial or portrait aesthetics, with a number of works treating the aesthetic problem generically [8–10] or categorically depending on its content [11] or query [12].

Motivated by these developments, we are inspired to improve existing deep learning techniques to better evaluate facial aesthetics. In this paper, we present a new compositional-based augmentation scheme for deep convolutional neural networks (DCNN) that can boost the aesthetic evaluation of facial portraits. Conventional sampling of sub-images (i.e. warping, center-crop, random crop) as receptive fields of a DCNN is rigid and can potentially distort the composition in portraits (Figure 1). As such, we exploit the rule of thirds composition rule to generate a feasible set of inputs to the network, which are in turn aggregated to form a deeply-learned augmented feature set. On a portrait-based dataset derived from the large-scale AVA dataset, our approach is able to surpass the performance of handcrafted and compositional-based features [3, 13] which have been a staple for defining beauty in portraits. The objective of this paper is to provide the methods of evaluating the portraits aesthetics using deep neural networks.

The following sections are organized as follows: Section 2 reviews the current works in literature, Section 3 describes the dataset that was used, Section 4 elaborates on the proposed method in detail, Section 5 reports the experimental results with further discussion on them, while Section 6 concludes the paper.

2 Related Work

The task of evaluating the aesthetic quality of photographs is becoming increasingly popular in recent years. As the notion of aesthetics is a subjective one, substantial human effort to rate photos is needed. Nonetheless, the creation of various datasets such as CUHKPQ [11] and AVA [9], from online photo communities such as Photo.net and DPChallenge [14], have made it possible to evaluate

solutions with ground truth labels. Works from the early efforts of Datta et al. [7] to the recent use of popular descriptors [8] have concentrated on the extraction of hand-crafted visual features for aesthetic evaluation of photos. In [7], as many as 66 pre-determined features, depicting colourfulness, composition, textures and statistical properties were crafted while [8] went for generic representations such as SIFT and Color descriptors, supplemented by Bag-of-Words (BoW) and Fisher Vector (FV) encodings and spatial pyramids (SP).

Different forms of convolutional neural networks [15, 10, 16] have been applied to solve this task, with the intent of learning features in a more natural and unguided manner. We take particular interest in the work of Lu et al. [10], which generated two heterogeneous inputs to represent global and local cues respectively. Global views are represented by normalized inputs such as center-crop, warp and padding, while local views are represented by randomly-cropped inputs of smaller parts of the image. These views were used generically for any type of photo, and may not be suitable for photos with facial portraits. However, a majority of DCNN-based works showed scant attempts to consider compositional impact relating to photographic rules.

However, not many works focused on the aesthetics of photos from a specific domain, particularly in facial portraits. The works in [3, 13, 17] try to rate the aesthetic quality of face portraits and understand the visual features that define the beauty of digital portraits, while another recent work [18] attempted to answer the question of "how to take a good selfie?" from the perspective of image popularity.

Khan and Vogel [3] made a comparative study on the use of traditional features [7] versus their choice of features that are focused on spatial and color composition. Firstly, template based spatial composition features were constructed from three templates for power points, power lines and their combination which were modified from [19]. Secondly, highlight and shadow composition features were obtained by extracting the face illumination, background contrast, brightness and size of the face as features. Their results show that the proposed features are more informative, more concise and better equipped to rank portraiture composition. The proposed features are more accurate by 6% and smaller 9 times in feature length compared to traditional features. More importantly, the authors postulated that spatial composition is more important than highlight and shadow composition when identifying aesthetics.

Redi et al. [13] introduced their own features to describe the portrait aesthetics. These features have been derived from the following photographic dimensions: compositional rules (lighting, sharpness, traditional compositional features), semantics and scene content (object bank feature), basic quality metrics (noise, exposure quality, JPEG quality), portrait specific features (facial landmark sharpness, landmark statistics and face/background contrasts) and fuzzy properties (emotion, originality, memorability and uniqueness). Their feature analysis showed that race, gender and age of a person are uncorrelated with photographic beauty, while soft and hard image properties such as sharpness of facial landmarks, image contrast, exposure, uniqueness and originality are

directly related to portrait aesthetics. The proposed method (after shrinking to a selected set of 12 positively-correlated features) outperformed the baseline method of [3] by 14.66% on the same data. However, using only face-related features without any selection, it performed better by only ∼1%. Another work by Lienhart et al. [17] applied similar types of features on three small datasets, while also performing aesthetic score prediction by support vector regression. But their work does not have the similar rigor and extensiveness as that of [13].

3 Dataset

The Aesthetic Visual Analysis (AVA) [9] dataset is the largest known dataset that is widely used to evaluate the aesthetic value of the images. The dataset contains more than 250,000 images from DPChallenge [14] with ground truth data such as human ratings ranging from 1 to 10 (with average 210 ratings per photo), with further meta-annotations from 66 textual semantics and 14 photographic styles from 72 different photo challenges. Following the work in [13], we select from the entire AVA dataset photos that are tagged with "Portraits" and also those whose challenge title contained the words Portrait, Portraiture or Portraits. Then, we filter further by detecting for photos that contained the presence of one or more faces with Face++ API [20]. A total of 10,141 images were obtained to form the portrait subset dataset. For better clarity, we refer to this subset as "AVA-Portraits" from this point onwards.

The same evaluation methodology as [13] was used. The dataset is randomly divided into five equal partitions. To ensure fair experimentation, we obtained the partition information from the authors. One partition will be used as test set and the rest as training set. Using 5-fold cross validation, the final accuracy is determined by averaging the minimum and maximum result of these five different cases. A portrait image is classified as aesthetically high if its aesthetic score is more than 5.5 (mean user score) [13]. Otherwise, the image is classified as aesthetically low.

4 Proposed approach

Convolutional neural networks (CNN) are biologically-inspired variants of the multilayer perceptron, dating back to the work of LeCun et al. [21]. However, it started to be increasingly popular only in 2012, after the work of Krizhevsky et al. [4] introduced an eight-layer deep CNN network, winning the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012. From then on, convolutional neural network models have shown unrivaled performance in ILSVRC each year. In ILSVRC, models are trained and tested on the ImageNet dataset [22], which consists of more than 14 million of images with 1,000 classes. The classes comprise of a large variety of objects such as car, motocycle, person, cats, etc. Realistically, an immense amount of computational power is necessary to be able to train weights in these deep networks at a reasonable amount of

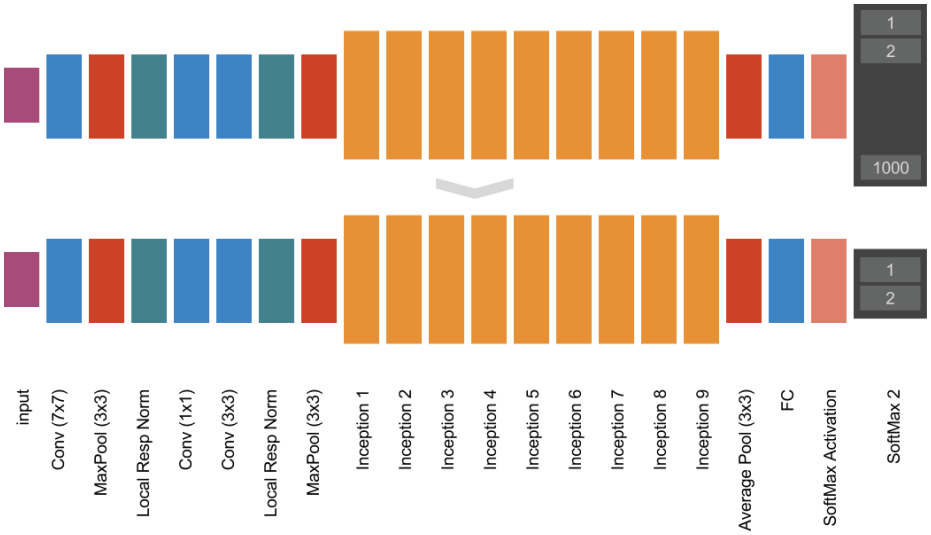

Fig. 2. Changing the number of nodes in the last layer of GoogleNet.

time. Thus, pre-trained models are commonly available for application to different related classification problems. However, in view of the different number of classes in other problems, the number of nodes in the last softmax layer should be replaced with that equivalent to the number of classes. In our case, the softmax layers are 2x1 as it is a binary classification problem (good aesthetics, bad aesthetics) (See Figure 2).

There are a number of popular pre-trained models available for public use. In this work, we use the Overfeat [5] (winner of the localization task in ILSVRC 2013) and GoogleNet [6] (winner of the classification task in ILSVRC 2014). For each of these models, the number of nodes in last (softmax) layer is reduced from 1000 to 2, and the weights are fine-tuned with the new training data. The second last layer of the network is then taken as the feature vector for classification by a softmax classifier (or multi-class binary logistic regression). We use the Sklearn-theano [23] library, an open source library on the Theano framework for the implementation of this research.

4.1 Feature augmentation using photographic rules

Feature augmentation is a process intended to increase the feature vector length at a certain layer by aggregating features obtained from various transformed versions of the input image. We distinguished this from *data augmentation*, which is a well-known technique [24, 4] used especially for learning deep representations by generating additional number of training samples. By augmentation we mean perturbing an image by transformations (e.g. cropping, flipping, or a combination), leaving the underlying class unchanged.

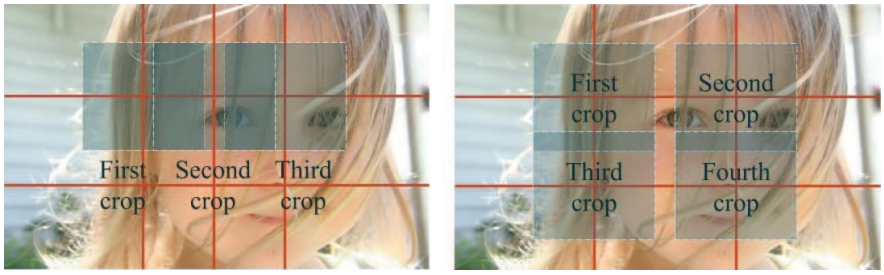

Fig. 3. a) Three-crop by rule of thirds, b) Four-crop by rule of thirds.

Fig. 4. a) Five-crop: four crops from four different corners + center crop) b) Five-crop: four crops from four corners + face crop.

In our work, we propose a feature augmentation scheme based on compositional photographic rules. This is motivated by the fact that typical image transformations such as cropping and flipping do not make aesthetic sense when it comes to face portraits. They may aggravate the actual composition of the portrait since a majority of cropping are done randomly or without applying compositional principles, while warping can potentially distort the face aspect ratio. The rule of thirds [19, 2, 25] is one of the popular photographic rules which divides the image to nine parts by 3 x 3 (see Figure 3). The four intersections of the rule of thirds lines are called power points. According to this popular rule, a photo is considered aesthetically appealing if the object of interest is located near one of these four power points.

We design a number of compositional cropping arrangements that are motivated by photographic rules associated to rule of thirds. Firstly, the baseline transformation involves warping the image to a constant width and height (corresponding to the model) and passing it as a single input to the DCNN model. We denote this in our experimental reporting as '**warp**'.

According to [3], the face, which is the main subject of interest in the portrait, should be located at the top power line by keeping the eyes in the upper third of the image area. Meanwhile, the symmetrical composition rule states that it is aesthetically pleasing to center the face in the image frame. Inspired

Fig. 5. Sample cropped faces detected by Face++ API

by these photographic composition rules, we modify the rule of thirds for portraits by taking two crops centered on the top two power points (left and right) and a center crop between these two crops (see Figure 3a). All three crops are passed to the DCNN model as individual inputs to generate feature vectors that are then concatenated. We denote this technique as '**c3**'. The third technique (denoted as '**c4**') uses four different sub-images of constant height and width, center cropped from four power points based on the rule of thirds (see Figure 3b). The augmented feature vector is formed by concatenating the four feature vectors generated from each crop.

Extending on to five-crop method, we propose two different variants. The first (denoted as '**c5**') takes four sub-images from the four corners of the image, and the center crop as the fifth. Since the fifth crop provides information from the middle of the photo, the use of four crops from the corners yields better coverage than using crops centered at all four power points. The second variant (denoted as '**c5f**') crops the face region as the fifth crop. Face++ [20] API was used to detect the location of the center coordinate, and a bounding box that defines the sub-image is cropped and squashed to a constant width and height. Figure 5 shows some sample face regions detected by Face++ API and used as the fifth crop for method '**c5f**'. Similar to previous schemes, DCNN features resulting from all five crops are concatenated. Due to slight unreliability of the Face++ API, only about 77.24% of all faces from the AVA-Portraits dataset can be detected. As such, the affected images fall back on the center crop as the fifth crop. Alternatively, we also manually annotated the faces in the remaining affected images. We denote this as '**c5f-m**'.

For all compositional cropping methods, we crop the sub-images based on a fixed width and height of 256 pixels, which are subsequently resized to receptive field sizes of their respective networks i.e. 234 for Overfeat, 224 for GoogleNet, both in RGB color space with zero mean.

Table 1. Aesthetic classification accuracy and dispersion error of the proposed methods compared against baseline methods.

Methods	Accuracy (%)	Error ± (%)
Baseline A [3]	54.585	1.25
Overfeat-warp	63.211	1.48
GoogleNet-warp	63.751	0.85
Overfeat-c3	63.861	0.44
Overfeat - c4	64.133	0.62
Baseline B [13]	64.240	1.76
GoogleNet-c3	64.244	0.64
GoogleNet-c4	64.519	1.06
Overfeat-c5	64.536	1.11
GoogleNet-c5	64.962	1.20
GoogleNet-c5f	**65.216**	1.20
GoogleNet-c5f-m	**65.330**	1.01

5 Results and Discussions

5.1 Overall results

We define two baselines for the purpose of comparisons– one by [3] which relies on traditional spatial and color composition features ('**Baseline A**'), and another [13] which uses a discriminative set of visual features ('**Baseline B**'). Two DCNN models were utilized in our experiments: the 8-layer **Overfeat** model and the 22-layer very deep **GoogleNet** model.

Table 1 shows the proposed and baseline approaches on the AVA-Portraits dataset. We achieve the best result reported thus far for this dataset, of 65.33%. From the feature augmentation results, we can see that the aesthetic classification accuracy generally increases as more crops were added to augment the DCNN-learned features (see Figure 7a). The best performing scheme uses five crops, with the variant that takes a face as fifth crop marginally higher than the one with the center crop. Manual annotation of the faces (that were not detected by Face++) resulted in further improvements. This feature augmentation scheme showed that having the face as foreground and the four other crops as background provides sufficient representation to mimic the foreground and background contrast rule raised in [3]. Figure 6 shows some sample images that were correctly classified (true positive, true negative) and incorrectly classified (false positive, false negative).

Meanwhile, we observe that the GoogleNet model outperforms the Overfeat model for each feature augmentation approach (see Figure 7b). This is expected of a more contemporary DCNN architecture that is deeper and had previously yielded better results at the ILSVRC. Of course, this comes at the expense of heavier computational cost. Figure 8 shows the breakdown of classification performance into the the five parts as specified in [13]. There is no monopoly

Fig. 6. Classification result of some images from the AVA-Portraits dataset using the GoogleNet-c5f-m method. The four quadrants show sample images based on their actual ground truth aesthetic class and predicted aesthetic class (positive side indicates high aesthetic quality, negative side indicates low aesthetic quality).

by any of the competing methods in the proposed scheme while the warping technique is noticeably poor.

Conventional single image warping produced quite poor results, much worser than Baseline B [13] which relies on an eclectic bunch of features that are both handcrafted and learned. It is important to point out that DCNN based methods do not required careful selection of features [3], or needing to eliminate features that are poorly correlated [13]. This itself gives an immense advantage towards methods that are able to learn good features on it own with minimal intervention.

5.2 Influence of small faces

We made an interesting observation that the AVA-Portraits dataset contains so-called "portrait" images with small faces inside (see Figure 10 for some samples). This may be caused by the fact that photo challenges in DPChallenge [14] are based on themes that can be subjectively interpreted by participants (as they deemed fit), or that these photos could exemplify certain artistic qualities associated to portraiture. Hence, this is the reason why some photos are not captured closed-up, frontal, or from portrait-like angles.

We identified images with "small faces" by considering faces that have a bounding box area of smaller or equal to 49 pixels[1]. We investigated if small faces can possibly influence the overall aesthetics of these portrait images. The

[1] This size was roughly determined by observation. Most of the other images appear to have substantially larger faces.

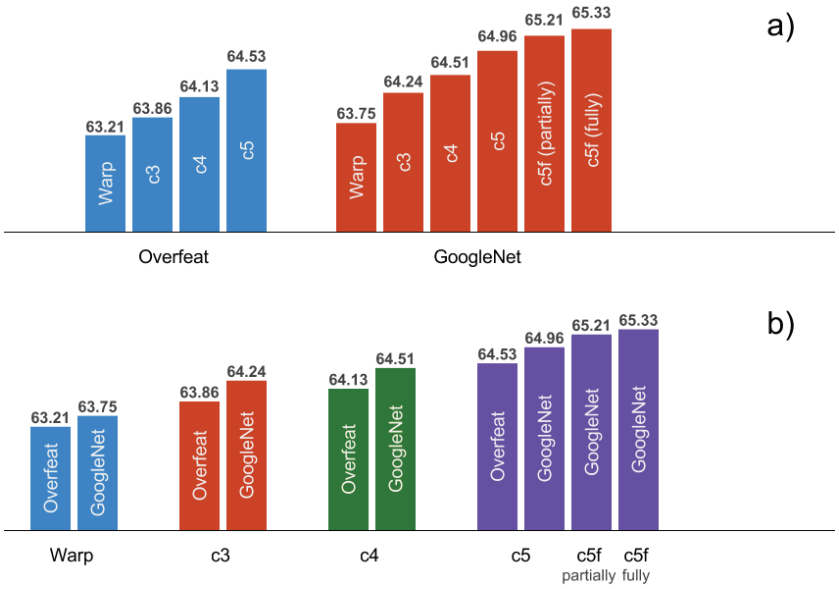

Fig. 7. Aesthetic classification accuracy grouped by (a) deep convolutional neural network models, and (b) type of feature augmentation scheme.

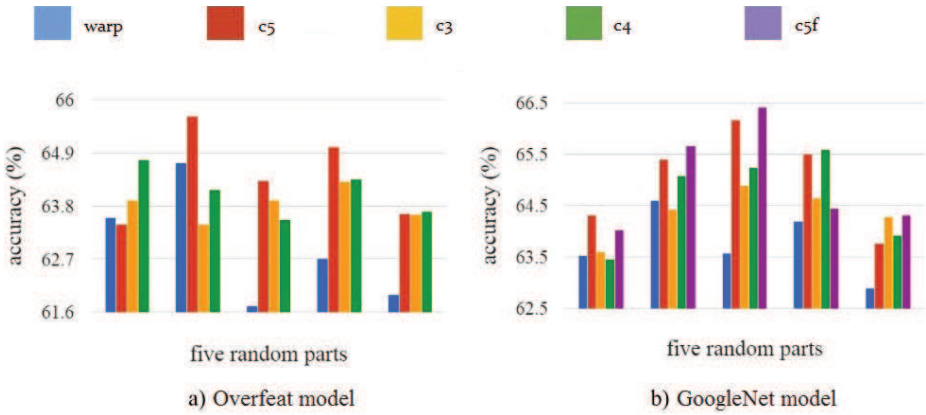

Fig. 8. Classification accuracy of the five random parts of the AVA-Portraits dataset using Overfeat and GoogleNet models.

whole dataset, and likewise, each of the five parts, contain approximately 17% of small face portrait images (Figure 9). So that we can conclude that small face portrait images, distributed almost evenly across the five parts. A total of 1,388 small face portrait images were found while the remaining 6,445 images contain normal size faces (those not marked by Face++ were ignored). Two sets of experiments were conducted on both small and normal size faces using the

Table 2. Comparison of classification performance between methods using small faces and normal faces on the GoogleNet-warp and GoogleNet-c5f methods.

Methods / Parts (%)	1st	2nd	3rd	4th	5th	Mean	Error
GoogleNet-warp-sf	65.10	62.96	63.12	62.16	59.91	62.50	2.59
GoogleNet-warp-nf	61.77	65.37	62.85	65.35	63.58	63.57	1.80
GoogleNet-c5f-sf	62.08	65.31	66.66	64.86	64.04	64.37	2.29
GoogleNet-c5f-nf	63.14	65.53	65.75	66.18	64.37	64.66	1.51

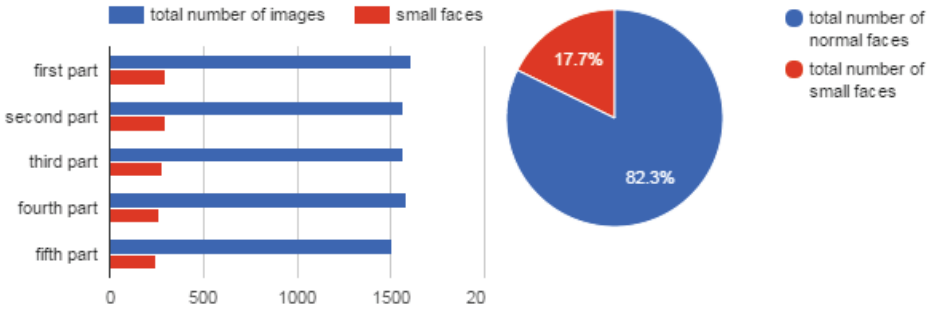

Fig. 9. Proportion of small faces in AVA-Portraits dataset

GoogleNet model. The first experiment uses the standard single image warping '**warp**', while the second uses the '**c5f**' five crop with face scheme. We indicate the faces sizes with '**nf**' for normal size and '**sf**' for small size.

Table 2 reports the accuracy of these evaluated methods for each of the five parts, and their respective mean accuracy and dispersion error. We observe that the photos with small faces performed slightly poorer than their normal face counterparts, with a larger spread of results. Considering that the proportion of small faces are evenly divided in all parts (Figure 9), the increase in dispersion error could mean that there are other non-face related aesthetic elements (such as image condition, background context, artistic styles, etc.) found in these portrait images that could potentially influence their actual ratings. Small faces are also likely to not contribute any meaningful impression of beauty to the viewer, in contrast to the way how close-up portraits do.

6 Conclusion

In conclusion, we introduce a new compositional-based augmentation scheme for deep convolutional neural network (DCNN) models to evaluate the aesthetic quality of facial portraits. Photographic rules relating to composition principles are adapted for portraits with the purpose of enriching inputs to the network with more visually pleasing crops. This way, we are able to generate a deeply-learned augmented feature set that can improve classification. On the AVA-

Fig. 10. Samples where small faces are found in "portrait" photos

Portraits dataset (a subset of the AVA dataset first used in [13]) consisting of more than 10,000 portrait-like images, our approach demonstrated an improvement in aesthetic classification over handcrafted baselines and standard DCNN models, achieving a top accuracy of 65.33%. In future, we intend to extend this idea by learning the compositional principles in facial portraits directly from the deep learning model itself. Also, we aim to explore other photo composition concepts besides the rule of thirds, such as visual balance and diagonal dominance.

References

1. Bakhshi, S., Shamma, D.A., Gilbert, E.: Faces engage us: Photos with faces attract more likes and comments on instagram. In: Proceedings of the 32nd annual ACM conference on Human factors in computing systems, ACM (2014) 965–974
2. Krages, B.: Photography: The art of composition. Skyhorse Publishing, Inc. (2012)
3. Khan, S.S., Vogel, D.: Evaluating visual aesthetics in photographic portraiture. In: Proceedings of the Eighth Annual Symposium on Computational Aesthetics in Graphics, Visualization, and Imaging, Eurographics Association (2012) 55–62
4. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. In: Advances in neural information processing systems. (2012) 1097–1105
5. Sermanet, P., Eigen, D., Zhang, X., Mathieu, M., Fergus, R., LeCun, Y.: Overfeat: Integrated recognition, localization and detection using convolutional networks. arXiv preprint arXiv:1312.6229 (2013)

6. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., Rabinovich, A.: Going deeper with convolutions. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. (2015) 1–9
7. Datta, R., Joshi, D., Li, J., Wang, J.Z.: Studying aesthetics in photographic images using a computational approach. In: European Conference on Computer Vision, Springer (2006) 288–301
8. Marchesotti, L., Perronnin, F., Larlus, D., Csurka, G.: Assessing the aesthetic quality of photographs using generic image descriptors. In: 2011 International Conference on Computer Vision, IEEE (2011) 1784–1791
9. Murray, N., Marchesotti, L., Perronnin, F.: Ava: A large-scale database for aesthetic visual analysis. In: Computer Vision and Pattern Recognition (CVPR), 2012 IEEE Conference on, IEEE (2012) 2408–2415
10. Lu, X., Lin, Z., Jin, H., Yang, J., Wang, J.Z.: Rapid: rating pictorial aesthetics using deep learning. In: Proceedings of the 22nd ACM international conference on Multimedia, ACM (2014) 457–466
11. Tang, X., Luo, W., Wang, X.: Content-based photo quality assessment. IEEE Trans. on Multimedia **15** (2013) 1930–1943
12. Tian, X., Dong, Z., Yang, K., Mei, T.: Query-dependent aesthetic model with deep learning for photo quality assessment. IEEE Transactions on Multimedia **17** (2015) 2035–2048
13. Redi, M., Rasiwasia, N., Aggarwal, G., Jaimes, A.: The beauty of capturing faces: Rating the quality of digital portraits. In: 11th IEEE Int. Conf. and Workshops on Automatic Face and Gesture Recognition (FG). Volume 1. (2015) 1–8
14. : Dpchallenge - a digital photography contest. (http://www.dpchallenge.com/) Retrieved: 26 August 2016.
15. Wang, Z., Dolcos, F., Beck, D., Chang, S., Huang, T.S.: Brain-inspired deep networks for image aesthetics assessment. arXiv preprint arXiv:1601.04155 (2016)
16. Lihua, G., Fudi, L.: Image aesthetic evaluation using paralleled deep convolution neural network. arXiv preprint arXiv:1505.05225 (2015)
17. Lienhard, A., Ladret, P., Caplier, A.: Low level features for quality assessment of facial images. In: 10th Int. Conf. on computer Vision Theory and Applications, VISAPP. (2015) 545–552
18. Kalayeh, M.M., Seifu, M., LaLanne, W., Shah, M.: How to take a good selfie? In: Proceedings of the 23rd ACM international conference on Multimedia. (2015) 923–926
19. Obrador, P., Schmidt-Hackenberg, L., Oliver, N.: The role of image composition in image aesthetics. In: IEEE Int. Conf. on Image Processing. (2010) 3185–3188
20. : Face++: Leading Face Recognition on Cloud. (http://www.faceplusplus.com/) Retrieved: 26 August 2016.
21. LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. Proceedings of the IEEE **86** (1998) 2278–2324
22. : ImageNet Database and Challenge. (http://www.image-net.org/) Retrieved: 26 August 2016.
23. : Sklearn-theano library. (https://sklearn-theano.github.io) Retrieved: 26 August 2016.
24. Chatfield, K., Simonyan, K., Vedaldi, A., Zisserman, A.: Return of the devil in the details: Delving deep into convolutional nets. arXiv preprint arXiv:1405.3531 (2014)
25. Child, J.: Studio photography: Essential skills. CRC Press (2013)